

Optimal Resource Allocation for Distributed Descriptor Computation

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1. Introduction

Calculation of molecular descriptors is a significant component of cheminformatics (Todeschini & Consonni, 2010; Mauri et al., 2017) as it converts chemical structures into numbers which enable similarity analysis, virtual screening and machine learning-based drug discovery (Lionta et al., 2014; Walters et al., 1998; Vamathevan et al., 2019). These descriptions are increasingly becoming costly to obtain on a computer since the chemical libraries of modern times are continually growing both larger and complex. This decreases efficiency in processing. Distributed cloud computing allows more than one cluster node to execute programs simultaneously (Rocklin, 2015; Armbrust et al., 2010) and thus is an ideal approach to handling large chemistry datasets. However, experience has shown that the execution time does not drop monotonically with the addition of computational resources, instead, in the initial stages, the performance is improving with increasing parallelism, but at some stage of time, the performance is deteriorating due to overheads in communication, synchronization, and cluster management (Amdahl, 1967; Dean & Ghemawat, 2008). Consequently, there is no way, cheaply or quickly, to experiment with more nodes to get the optimal selection of automated workflows. The paper attempts to identify means of enhancing the distributed cheminformatics apps through developing a method to predict the time that it may take to execute something, depending on the size of the dataset, as well as the cluster configuration. This orientation simplifies the calculation of the number of nodes that will be required to achieve maximum performance, efficiency and scalability and reduces the time of execution to a minimum.

2. Related Work and Motivation

To conduct the experiment and observe the impact of dataset size and cluster design on the time required to complete the calculations, it was highly essential in previous research related to the computation of distributed molecular descriptors (Didachos et al., 2022). These studies showed that the optimal cluster size was possible by executing the same workload using the various node configurations and selecting the one that had the least runtime. Nevertheless, the latest studies have indicated that the cloud and parallel models have the potential to hugely accelerate the extraction of descriptors (Didachos et al., 2023) (Lavecchia, 2015; Mak & Pichika, 2019). Although this approach is practical, it is computationally intensive and cannot be used when the size of datasets or system parameters varies since the trials should be done on all possible occasions. Since it cannot be applied in other conditions, empirical tuning lacks generalizability hence cannot be used for automated decision-making or real-time decision-making in the dynamic cloud world. These limitations motivate the researchers abandon empirical optimization in favour of analytical models. It is a math-based approach that incorporates real-life performance data in order to create a direct forecast of the manner in which resources will be distributed. This reduces redundant expenses, enhances scalability and makes distributed computing processes more efficient and repeatable.

3. Experimental Setup and Data Collection

The experiment aimed to determine the influence of cluster configuration and size of the dataset on the time required to execute a distributed molecular descriptor calculation. Descriptor extraction is directly associated with the complexity of the molecules (Kombo et al., 2013; Todeschini & Consonni, 2010) in its cost. The records consisted of chemical compounds (SMILES strings) with an average length of 57 characters. Each sample had low, medium and

high workloads of between 5,000 and 50,000 chemicals. We undertook the primary research on a platform of distributed computing based on cloud computing and cluster architecture. The descriptor computing job was being executed concurrently by a network of nodes of varying numbers of nodes. The cluster arrangements of 25 to 185 nodes allowed the examination of the effects of performance scaling and overhead in more detail. The specific time required to run each dataset was written down against each cluster design. This involved forming the cluster, scheduling, distributing data, inter-node communication, ensuring that everything is synchronized and calculating descriptors. The results accurately represented real distributed execution behaviour rather than idealized computing performance due to the comprehensive measure technique that we adopted.

Table 1 Execution Time (sec) for a variety of Cloud Infrastructures 1/2

Number of Nodes	5K rows	10K rows	15K rows	20K rows	25K rows
25	169,55	211,74	371,77	476,84	568,92
35	171,38	212,75	288,31	288,19	397,3
45	170,92	211,78	209,89	250,09	382,3
55	151,84	199,87	200,64	239,43	296,24
65	133,91	187,78	180,44	238,09	251,37
75	130,62	165,74	183,49	236,41	250,26
85	134	155,7	215,73	238,37	254,56
95	137,81	158,37	189,36	217,29	255,59
105	142,98	155,15	176,36	198,76	253,42
115	147,65	155,93	179,09	195,91	253,2
125	152,56	152,81	198,74	192,62	242,59
135	154,38	153,03	193,77	187,45	236,16
145	140,89	152,2	184,07	183,19	228,92
155	146,92	161,05	186,25	201,25	229,86
165	145,26	170,19	178,08	211,94	222,94
175	149,1	170,79	176,77	216,76	214,26
185	159,8	170,61	178,14	220,62	212,91

Table 2 Execution Time (sec) for a variety of Cloud Infrastructures 2/2

Number of Nodes	30K rows	35K rows	40K rows	45K rows	50K rows
25	469,3	829,06	607,89	1043	794,42
35	467,2	533	500,8	637,9	706,5
45	458,55	361,52	395,54	414,51	681,63
55	389,11	358,62	389,12	413,23	580,6
65	362,46	355,71	386,91	408,91	435,98
75	290,99	314,75	368,41	407,4	429,4
85	259,93	297,48	307,4	409,1	383,37
95	256,14	295,27	295,36	396,36	357,57
105	246,72	269,87	280,3	378,58	349,17
115	249,75	269,88	279,69	378,69	343,35
125	249,03	265,58	263,66	382,69	328,69
135	251,47	265,41	360,07	315,47	319,07
145	248,68	263,3	251,3	297,45	315,45
155	253,3	258,86	256,22	313,12	319,97
165	242,96	239,73	253,36	337,2	315,7
175	242,88	239,73	253,56	332,46	309,42
185	245,25	239,94	256,17	332,04	307,57

Table 1: Execution times (seconds) for varying dataset sizes and cluster configurations.

4. Mathematical Performance Modeling

Mathematical performance modelling explicitly relates computational resources to time to execute using variables of the experiment. N is the size of the cluster nodes that will be used to manage the program, and D is the size of the data or the quantity of chemical compounds to be processed. A performance model was developed as a regression model (Montgomery et al., 2012) to make a guess on the total execution time. It applied experimental measurements of execution time to various combinations of N (25 to 185 nodes) and D (5,000 to 50,000 compounds). This results in the formulation of the following model: $T(N,D)=a+bN+cD+dN^2+eND$ where $T(N,D)$ represents the total execution time, including both computation and distributed system overheads. The advantages of parallelism and computational workload involved in the calculations are reflected in the linear components' bN and cD , which are the first-order effects of increasing the size of the cluster and the data, respectively. The quadratic term dN^2 models the non-linear overhead introduced by excessive parallelism, such as the communication costs, synchronization costs, and scheduling costs, which increase too quickly with the addition of more nodes. Finally, there is an interaction term, eND which is used to determine the relationship between the size of the dataset and the cluster configuration. This demonstrates the influence of workload and the sharing of resources on scaling behaviour. Taken together, these terms enable the model to model empirically observed non-monotonic performance in the distributed computation of descriptors in terms of execution time.

5. Analytical Computation of Optimal Nodes

Assuming that the size of the dataset D remains constant, the performance model can be reduced to a single variable as a function of the number of cluster nodes N . It simplifies the process of straight analytical optimization. Replacing an execution time model with a constant number of D allows obtaining a convex quadratic function (Boyd & Vandenberghe, 2004) in N . Terms which are dependent on the data are represented as fixed coefficients in the function. The fewest number of nodes is through the derivation of T (distinct from D) as per N and establishing it as 0. This will reduce the time required to operate. The solution to this problem will provide us with the analytical formula, as far as the optimal number of nodes (or the number of nodes in a cluster that decreases the time taken to run a given dataset) is concerned, which is expressed as $N^*(D)$ which specifies the cluster size that minimizes execution time for a given dataset. When there is a positive quadratic coefficient, it is definite that the function is convex with reference to N . This implies that the equilibrium is a global minimum rather than a local extremum. The value $N^*(D)$ obtained through analysis is rounded to the nearest whole number, and capped with the minimum and the maximum number of nodes that the cloud infrastructure will be capable of supporting to ensure that it is useful and brings performance close to optimal in practice.

For a fixed dataset size D , the execution time model reduces to a single-variable function of the number of cluster nodes N :

$$T(N) = a + bN + cD + dN^2 + eND$$

To determine the optimal number of nodes that minimizes execution time, the partial derivative of $T(N)$ with respect to N is computed:

$$\frac{\partial T(N,D)}{\partial N} = b + 2dN + eD$$

Setting the derivative equal to zero yields the stationary point:

$$b + 2dN + eD = 0$$

Solving for N produces the analytical expression for the optimal number of nodes as a function of dataset size:

$$N^*(D) = -\frac{b+eD}{2d}$$

Since the constant d of the term N^2 is positive, the execution time functional is convex along the N . It implies that N^*D is not a local extremum of the execution time but a global minimum. In practice, the analytically determined value of $N^*(D)$ is rounded off to the number of nodes in a cluster that can be utilized. The reason behind this is to make things realistic without necessarily approaching optimum performance.

6. Experimental Validation of the Analytical Model

Each dataset size had its execution time plots created by measuring the overall time of molecular descriptor computation, in addition to varying the number of cluster nodes in a predetermined manner. The graph indicated the run time according to the number of nodes per dataset of between 5,000 and 50,000 chemicals. The resulting standard convex curves indicate the trade-off regarding the amount of parallelism and the additional workload of distributed systems. The processing time reduces initially with the addition of more nodes, but increases later due to the visibility of the costs of communication, synchronization and scheduling (Dean & Ghemawat, 2008; Armbrust et al., 2010).

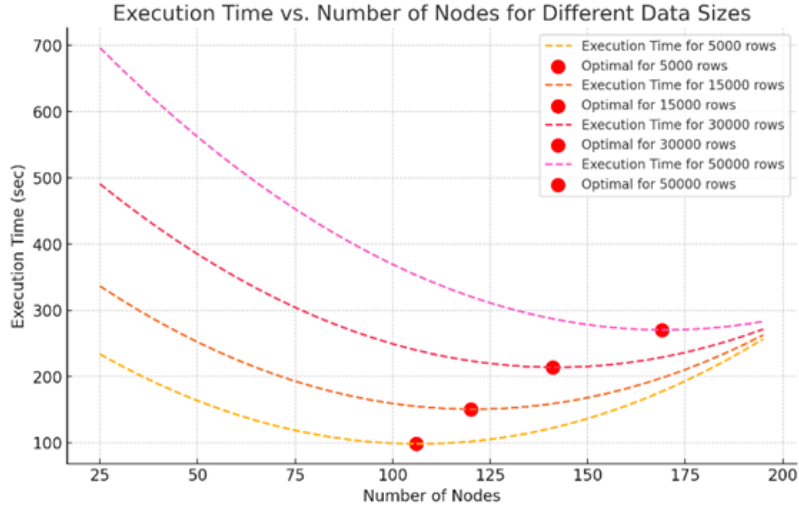


Figure 1: Execution time as a function of cluster size for different dataset sizes, showing experimentally observed minima.

In order to determine the experimental minimum of each experimentation time curve, we identified the node configuration with the fastest runtime for a particular dataset size. The empirical minima are the most successful cluster setups that the analytical model has discovered after extensive trial and error to serve as a benchmark for evaluating the analytical model. We should not rely on a systematic optimization approach, but rather on the fixed or heuristic-based resource allocation approach. This is due to the fact that the observed minima varied with the data set sizes, indicating that the optimum number of nodes varies with the data.

The lowest values obtained in experiments were compared to the optimal node values proposed by the analysis that were obtained using a mathematical performance model. The anticipated optimal number of nodes was quite similar to what was observed in all the sizes of data sets that were experimented with. The difference was only in minor amounts, which could be attributed to the rounding of the parameters to integers and the restriction of the infrastructure. This rigorous contract is a quantitative demonstration that the model is accurate, in that the

model of analysis can be used to recreate the behaviour of the real system, rather than trial-and-error testing, which is time-consuming and error-prone.

7. Performance Comparison: Initial vs Proposed Approach

The initial empirical resource selection process performed a significant number of experiments using varying numbers of nodes per dataset size to determine the optimum cluster designs. The initial method executed on a dataset containing 10,000 chemicals and an 180-node set required approximately 198.9 seconds to run. The selected configuration of 100 nodes used 263.6 seconds on a dataset of 20,000 compounds and used 445.3 seconds on a dataset of 50,000 compounds. It was a trial-and-error approach that did not always result in good utilization of resources, but made good combinations.

The proposed mathematical optimization approach does not presuppose that changes are made via trial and error; the optimal number of nodes is straightforwardly found with the help of the popular performance model. In 10,000 and 20,000 compounds, the model found 145 nodes in the analysis optimal, whereas in 50,000 compounds, the model found 185 nodes in the analysis optimal. Instead, we simply made these carefully selected configurations by just using the execution time function that we obtained through regression and the optimal solution of that solution, which was a closed form. We didn't do any more tests.

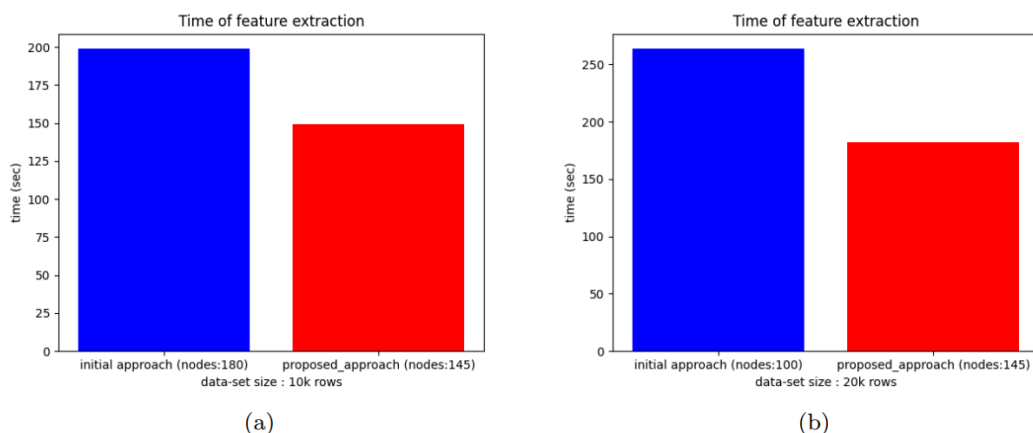


Figure 2: Comparison of execution time between empirical and analytically optimized cluster configurations.

A comparison between the times of implementation of various plans reveals that the proposed approach is better and quicker. The 10000-compound dataset showed that the 145-node architecture reduced the processing time to 152.2 seconds (compared to 198.9 seconds), and this was what theoreticians believed would occur. The execution time of the 20,000-compound dataset decreased to 183.1 seconds, as compared to the 263.6 seconds with greater nodes compared to the initial empirical selection. This is a significant improvement. The processing time of the 50,000-compound dataset using the 185-node cluster identified using analysis was reduced from 445.3 seconds to 307 seconds. These findings demonstrate that the proposed approach will always reduce the time of executions by utilizing computing resources in a more efficient manner. It also has better runtime as well as utilization of cloud infrastructure, which costs less.

8. Interpretation of Results

The consistently convex execution time curves that have been drawn based on dataset sizes 5000 to 50000 compounds indicate that there is an optimum number of cluster nodes with which to operate a given task. Processing time reduces significantly with the increase in the number of nodes. This implies that there is less work to be performed by each node and even more parallelism. When the number of nodes approaches the theoretically ideal value and the amount of time required to execute a task becomes minimum, adding nodes to the cluster has a decreasing impact on speed. This action demonstrates that the optimal method of selecting nodes varies with the size of the data set and cannot be located under fixed or heuristic settings.

The interaction between parallel computing and additional work associated with distributed systems brings the observed non-monotonic growth. Initially, node addition accelerates the description processing. Nevertheless, high amounts of parallelism have high costs in terms of cluster management, addressing timing problems, node-to-node communication, and work planning. Ultimately, those expenses rise disproportionately with the size of the cluster and exceed the value of additional working power. The regime of overheads becomes more pronounced as the cluster increases in size. Although additional nodes can be used, execution time increases, indicating that overprovisioning of resources can be detrimental to distributed cheminformatics jobs.

The automatic selection of resources is a research project that we are working on, and can assist cloud-hosted scientific methods significantly. To maximize the resources assigned, analytically developed performance models that are accountable for the task size and system overheads are superior to fixed specifications or experiential changes. Automated methods are able to reduce runtime, minimize the cost of the cloud, and enhance overprovisioning by

estimating the optimal cluster size prior to its utilization. This technology is most effective in situations where it is incorporated into automated workflow orchestration systems with applications in a wide variety of distributed computing environments (Gustafson, 1988; Montgomery et al., 2012).

9. Contribution and Generalization

The proposed approach is a great leap of improvement over what has already been accomplished, since it takes performance statistics of experiments to design a plan for predicting and analyzing the optimal utilization of resources in the distributed descriptor computing. Here, this paper outlines a regression-based execution time model that, in particular, computes the optimal number of nodes to be used for a specific workload by considering the influence of both dataset size and the architecture of the cluster rather than relying on mere observations. The approach provides an orderly and repeatable mechanism of selecting resources to be used in distributed cheminformatics applications through the solution of performance minima mathematically. This connects mathematical optimization and experimentation.

The proposed approach is, in many aspects, superior to the typical trial-and-error approach. It eliminates the need for repetition of experiments in various cluster setups by saving on setup time, processing overhead and involvement of users. The mathematical approach also ensures that one can immediately make an educated guess of the most efficient cluster sizes prior to their utilization, preventing the under- or over-provisioning of resources. Once subjective or ad hoc decision-making is considered out of the optimization, processing time and cloud computing costs are reduced in a quantifiable manner. It makes it easier to do again.

The creation and validation process was grounded on the computation of molecular descriptors. Nevertheless, the technology is versatile and can be applied in an extremely broad scope of scientific processes. A data-driven modelling and optimization method can be used for jobs which do not scale linearly due to the communication and synchronization costs. The proposed structure will result in execution being scalable and automated in numerous fields of science. It also provides a means of investing resources such that the performance in the cloud-based data processing, the high-performance computing and the large-scale analytics environment is maximized.

Conclusions

This contribution shows that the outcomes of the tests in the field of performance can be effectively converted to the analytical model to improve the use of cloud resources to calculate distributed molecular descriptors. The results also show that the correlation between the execution time and cluster size is not that simple. In its place, the number of nodes that analytically determine as optimal reduces the runtime of all data sizes. We show that an analytical optimization is a better and more stable way of optimizing to a distributed cloud environment than a trial and error by comparing the hypothetical optimum settings to the realised lowest settings. Such a strategy could be extended in the future to investigate the molecular complexity by examining the alterations in the SMILES length, to model the models that would be able to adapt to the varying workloads depending on multi-template or adaptive strategy and to generalize such a methodology with large scale cloud orchestration and scientific workflow automation systems that would be able to manage autonomously and performance sensitive resource management across the full range of distributed computing applications.

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